EXPT NO:3	**CLEANSE DATA BY HANDLING MISSING VALUES, OUTLIERS, AND INCONSISTENCIES USING TABLEAU AND POWERBI**
DATE: 10.07.2026	

PRE-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What are the potential impacts of missing values in a dataset, and what are some common strategies to address them in Tableau and Power BI?

Missing values can reduce accuracy, distort totals and averages, and create misleading visualizations. They may also cause calculations or relationships to fail. Common strategies include removing incomplete records, replacing missing values with mean, median, zero, or a business-approved value, and creating an “Unknown” category. In Tableau, calculated fields and filters can handle nulls, while Power Query in Power BI supports replacement, filling, and removal.

2. What is the importance of identifying and addressing outliers in a dataset? How can outliers affect data analysis and model performance?

Outliers are unusually high or low values that can strongly influence averages, trends, correlations, and forecasting results. Some outliers represent genuine business events, while others may come from data-entry or system errors. If ignored, they can produce misleading insights and reduce model accuracy. Analysts should investigate the cause and then retain, remove, cap, or transform the values based on business context.

3. What types of data inconsistencies might you encounter during data cleansing, and how can Tableau and Power BI help in resolving them?

Data inconsistencies may include duplicate records, spelling variations, mixed date formats, incorrect data types, inconsistent category names, and different measurement units. For example, “South,” “south,” and “SOUTH” may be treated as separate values. Tableau helps through grouping, aliases, calculated fields, and data-type corrections. Power BI uses Power Query to remove duplicates, replace values, standardize text, split columns, and correct formats.

4. How do you identify patterns in data using visualizations to spot missing values, outliers, and inconsistencies?

Visualizations make data-quality issues easier to notice. Bar charts can reveal blank or duplicated categories, while line charts expose sudden gaps, unusual spikes, or unexpected drops over time. Box plots and scatter plots help identify outliers, and heat maps can show concentrations of missing values. Filters, labels, and colour coding allow analysts to isolate suspicious records and determine whether they represent valid patterns or data errors.

5. What is the role of calculated fields in data cleansing, and how can they be used to handle missing values and outliers in Tableau and Power BI?

Calculated fields allow analysts to create cleaned values without modifying the original dataset. They can replace nulls, standardize categories, flag suspicious records, and identify or limit extreme values. In Tableau, functions such as IFNULL, ZN, and conditional IF statements are useful. In Power BI, DAX functions like COALESCE, ISBLANK, and IF can create reliable columns or measures for analysis.

IN-LAB EXERCISE:

OBJECTIVE:

To clean a given dataset (e.g., Superstore Sample in Tableau and Financials/Sales Data in Power BI) by identifying and resolving missing values, outliers, and inconsistencies using built-in tools, ensuring the dataset is accurate and ready for analysis.

PROCEDURE:

1. **Open the Tool and Connect Dataset** ○ Open Tableau Desktop , Power BI Desktop.
   - Load the built-in dataset (Sample - Superstore in Tableau / Financials in Power BI).
   - Preview the dataset to understand its structure.
2. **Initial Inspection** ○ Explore fields such as Sales, Profit, Region, or Product Category.
   - Check for nulls, inconsistent text values (e.g., "South Region" vs. "South"), or date formatting issues.
3. **Handling Missing Values** ○ Use Data Interpreter (Tableau) / Power Query (Power BI) to highlight nulls. ○ Replace missing numerical values with mean/median, and text with "Unknown".
   - Use calculated fields: IFNULL([Field], Default) in Tableau or IF(ISBLANK([Field]), "Unknown", [Field]) in Power BI.
4. **Detecting and Managing Outliers** ○ Create scatter or box plots for Sales and Profit to identify extreme values.
   - Choose strategies:
     - Filter them temporarily.
     - Apply caps using calculated columns.
     - Flag them with labels for review.
5. **Resolving Inconsistencies** ○ Use Group/Alias (Tableau) / Replace Values (Power BI) to standardize text.
   - Normalize data formats, such as date fields or currency units.

2. **Validation with Visuals**
   - Build dashboards with histograms, bar charts, and box plots to visually confirm improvements.

- Compare before/after cleaning stages for accuracy.

3. **Save the Cleaned Dataset** ○ Save as .twbx (Tableau) / .pbix (Power BI).
    - Optionally export cleaned data to .csv.

4. **Summary Dashboard**
    - Create a visual summary of the cleaning process, highlighting:
        - Fields corrected.
        - Nulls/outliers removed or adjusted.
        - Clean dataset metrics (record count, completeness %).

Experiment Dataset:

1. Tableau: *Sample - Superstore*
2. Power BI: *Financials* or *Sales Data*

How to Access:

Tableau:

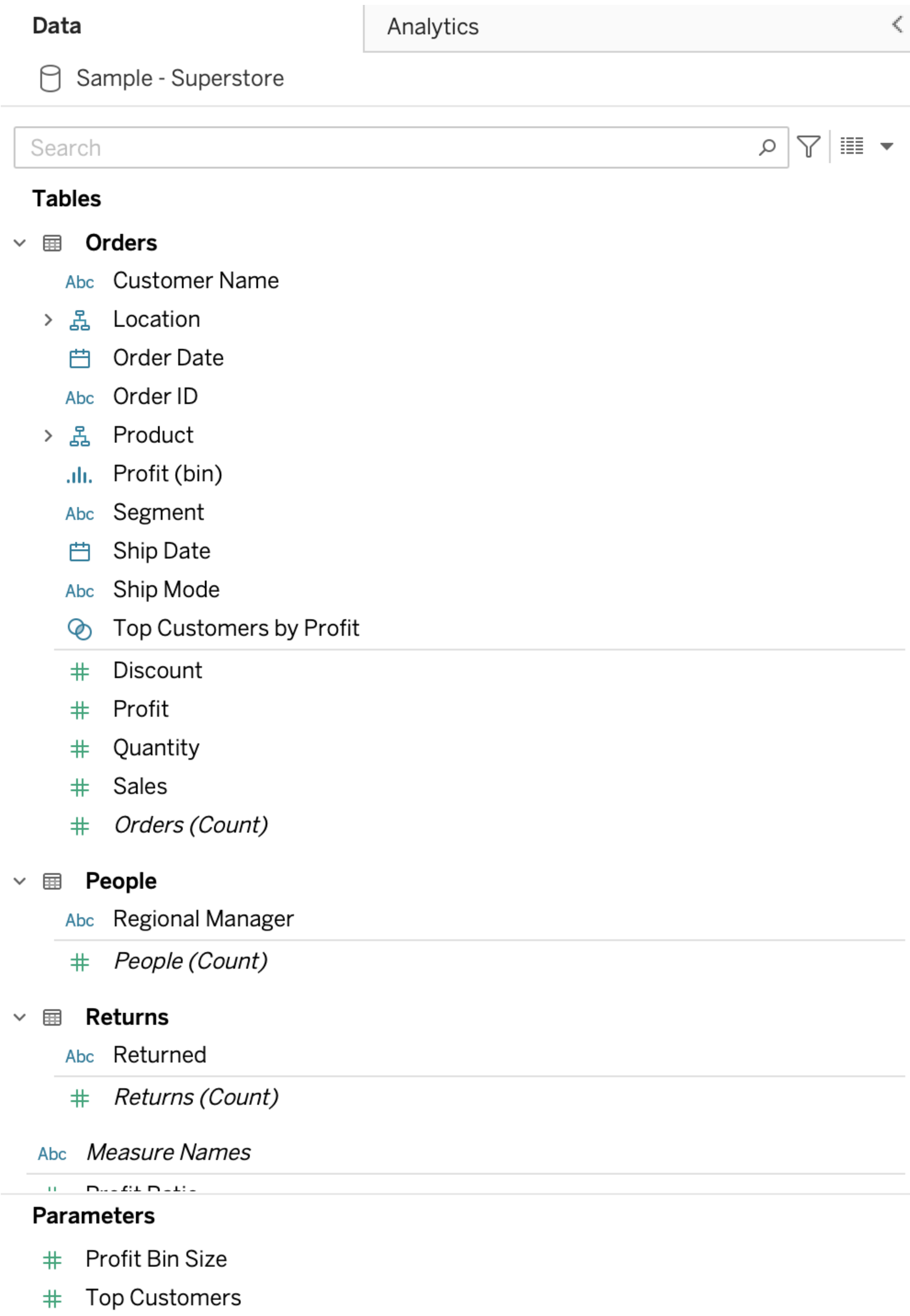
- **Navigate to:** *Start Page → Sample Workbooks*
- Or: *Help → Sample Data → Open Sample Data Folder*

Power BI:

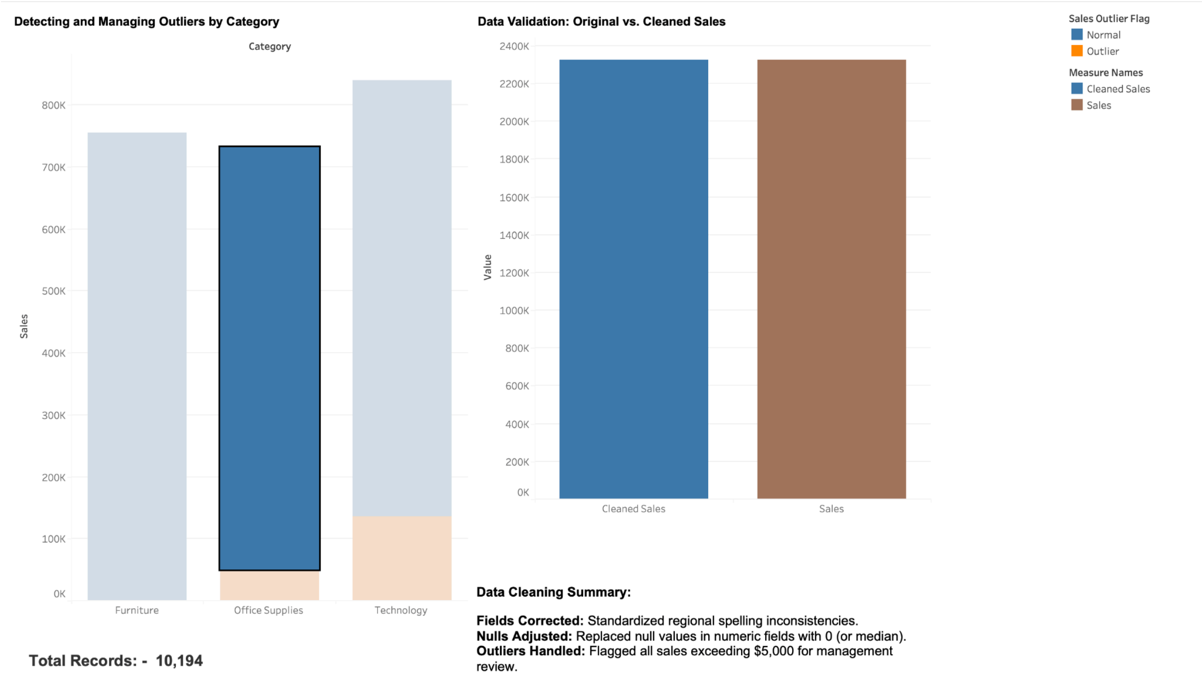
- **Navigate to:** *Home → Get Data → Excel → Choose sample datasets (e.g., Financial Sample.xlsx)*
- Microsoft also hosts Power BI sample datasets online.

Fields Used:

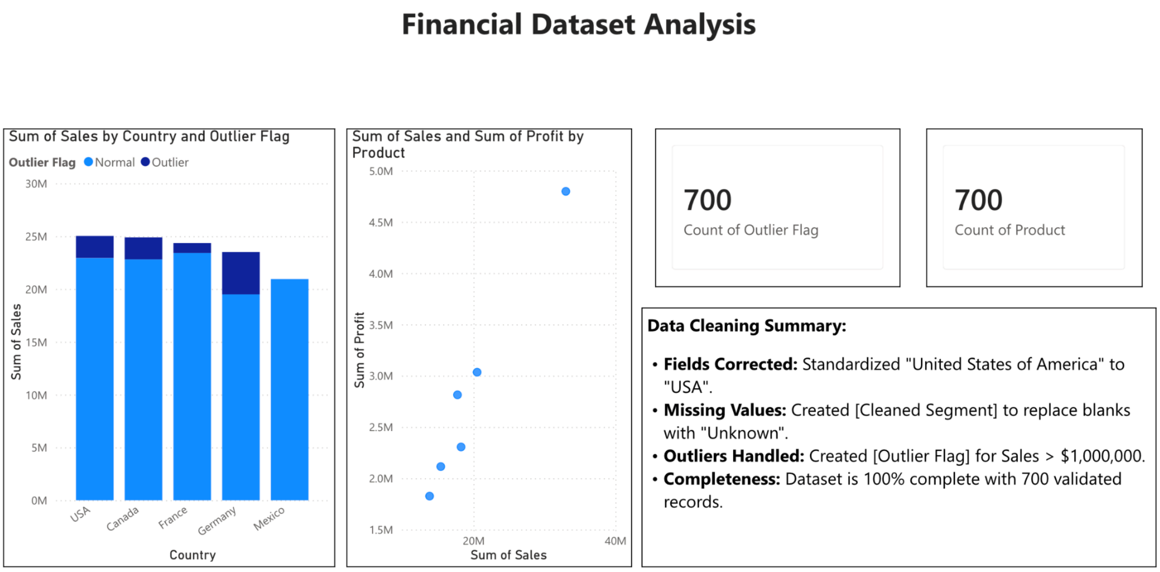
1. Sales
2. Profit
3. Region
4. Category
5. Order Date / Transaction Date



(INITIAL DATA INSPECTION)



PowerBI:



POST-LAB QUESTIONS (PROVIDE BRIEF ANSWERS TO THE FOLLOWING QUESTIONS)

1. What method did you use to replace missing values in numeric fields?

I replaced missing numeric values using the median because it is less affected by extreme values than the mean. In Tableau, I used calculated fields with functions such as IFNULL, while in Power BI I used Power Query to replace nulls. This approach preserved most records, reduced distortion in totals and averages, and ensured that charts and calculations remained complete and reliable.

2. How were outliers handled in your dataset and what visual confirmed their correction?

Outliers were first identified using unusually high or low values in box plots and scatter plots. After checking whether they were valid business cases or data errors, incorrect values were removed or capped using acceptable limits. The corrected box plot confirmed the improvement because the extreme points were reduced, and the spread of the data became more consistent without changing the overall pattern unnecessarily.

3. What inconsistencies were detected in categorical values and how were they standardized?

The categorical fields contained inconsistencies such as different capitalisation, spelling variations, extra spaces, and multiple labels for the same category. For example, "South," "south," and "SOUTH" appeared as separate values. I standardized them by trimming spaces, correcting spelling, and converting values to a common format. Tableau grouping and aliases, along with Power Query replacement tools, helped create consistent categories.

4. Describe how calculated fields helped in resolving data issues.

Calculated fields helped create cleaned versions of existing columns without modifying the original data. I used them to replace null values, standardize category names, flag outliers, and classify records as valid or suspicious. In Tableau, conditional expressions were used, while Power BI used DAX formulas. These fields improved consistency and allowed the dashboard to use reliable values for totals, comparisons, filters, and performance indicators.

5. How did your dashboard demonstrate the effectiveness of the data cleaning process?

The dashboard demonstrated effective cleaning by showing complete charts, consistent category labels, fewer extreme values, and accurate summary indicators. Before cleaning, some visuals had blanks, duplicated categories, and unusual spikes. After cleaning, totals aligned correctly, filters worked smoothly, and trends became easier to understand. KPI cards, bar charts, and box plots provided clear evidence that data quality and analytical reliability had improved.

RESULT:

The dataset was successfully cleaned and prepared for accurate analysis using Tableau and Power BI. Issues such as nulls, inconsistencies, and outliers were identified and resolved, improving data quality significantly.

ASSESSMENT

Description	Max Marks	Marks Awarded
Pre Lab Exercise	**5**	
In Lab Exercise	**10**	
Post Lab Exercise	**5**	
Viva	**10**	
Total	**30**	
	Faculty Signature	